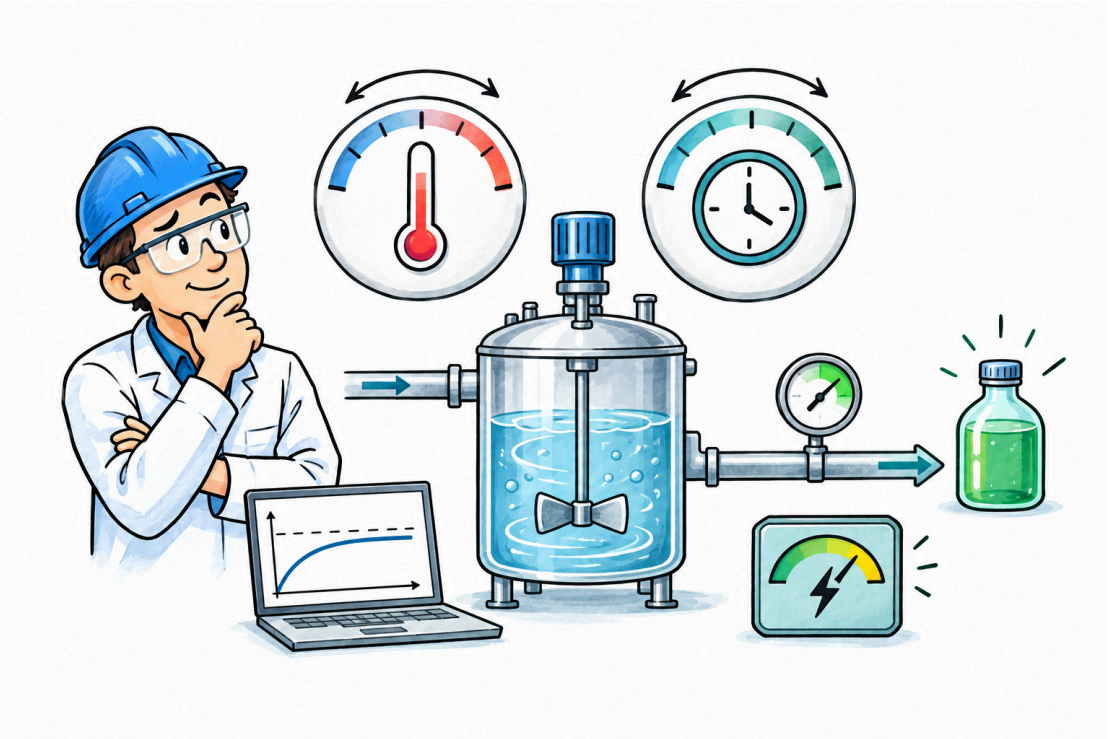
Week 1: Process Prediction and Surrogate Modeling Basics

Outlet concentration prediction at specified operating conditions

An engineer wants to predict outlet concentration and conversion at new temperature and residence-time settings.

Experiments or detailed process calculations take time at each condition. Could existing data help predict the response?



Prediction performance at new operating conditions

A model can reproduce its training data and still make large errors at new inputs. Units, operating range, and fixed assumptions also determine what its output means.

This week: prediction-task definition and independent evaluation. A small CSTR with a computable reference response lets us check each label.

Predictor fitting and independent evaluation

1. Input, output, context, units, and domain definition
2. Steady-state reference data and separate train, validation, and test sets
3. Quadratic regression baseline and test errors in physical units
4. Concentration-conversion consistency and locations of large errors

Course roadmap: Week 2 learns multicomponent MLPs; Week 3 adds time response; Week 4 examines physical consistency. Weeks 5-7 address optimization formulations and model structure. Week 8 selects and revalidates operating conditions.

Deliverable: predictor with an independent evaluation.

1. Define the prediction contract

Role	This example
Inputs	Temperature T [K], residence time τ [min], feed concentration C_{Af} [mol/L]
Outputs	Outlet concentration C_A [mol/L], conversion X [−]
Domain	300–360 K; 1–10 min; 0.8–1.5 mol/L
Fixed context	Ideal mixing, constant volume/density, first-order A → B, declared rate parameters
Reference	Steady-state solution of the specified CSTR balance

Feed concentration: a variable input in this prediction task.

One input row describes one operating condition. The same temperature with a different residence time or feed concentration asks a different question, so all three values travel together. The two outputs describe concentration and conversion after equilibration.

2. Reference CSTR model

Use an ideal, perfectly mixed, constant-volume, isothermal CSTR with a first-order reaction A → B, constant density, and no B in the feed. Reactor temperature T is an operating input.

$$\frac{dC_A}{dt} = \frac{C_{Af} - C_A}{\tau} - k\left(T\right)C_A$$

Here concentrations are in mol/L, time and residence time in minutes, T in kelvin, and k in inverse minutes. The flow-to-volume ratio is the reciprocal of residence time. The inlet-outlet term and the reaction term therefore both have units of mol/(L·min).

The tank contains V·C_A of A. Inflow Q·C_Af minus outflow Q·C_A and reaction consumption V·k·C_A gives the rate of inventory change. Divide by V and substitute Q/V=1/τ to obtain the balance above.

Temperature dependence and operating domain

Temperature-dependent rate constant:

$$k\left(T\right) = k_{\mathrm{ref}}\exp\left[\beta\left(\frac{1}{T_{\mathrm{ref}}} - \frac{1}{T}\right)\right]$$

Parameter or input	Value or range	Interpretation
Reference rate constant	0.2 per minute	Rate at the reference temperature
Reference temperature	330 K	Rate-law reference
β = E/R	6000 K	Synthetic temperature-sensitivity parameter
Temperature	300–360 K	Sampling and evaluation domain
Residence time	1–10 min	Sampling and evaluation domain
Feed A concentration	0.8–1.5 mol/L	Sampling domain; reference map at 1.2

At T=330 K the exponent is zero, so the exponential factor is one. Raising temperature decreases 1/T and increases the exponent and reaction rate. Use kelvin: inserting Celsius changes the reciprocal-temperature difference.

3. Derive the steady-state map

Set the time derivative to zero, multiply by residence time, and collect the terms containing outlet concentration. The resulting mapping and conversion are:

$$C_A = \frac{C_{Af}}{1 + k(T)\tau}, \quad X = 1 - \frac{C_A}{C_{Af}} = \frac{k(T)\tau}{1 + k(T)\tau}$$

This gives a three-input, two-output map: temperature, residence time, and feed concentration map to outlet concentration and conversion. In this first-order model, conversion is independent of feed concentration.

Steady state does not stop the reaction. Inflow, outflow, and reaction continue, but their net effect is zero and tank concentration stays constant. The denominator combines outflow, represented by 1, with reaction over the residence time, represented by kτ.

A steady-state calculation, step by step

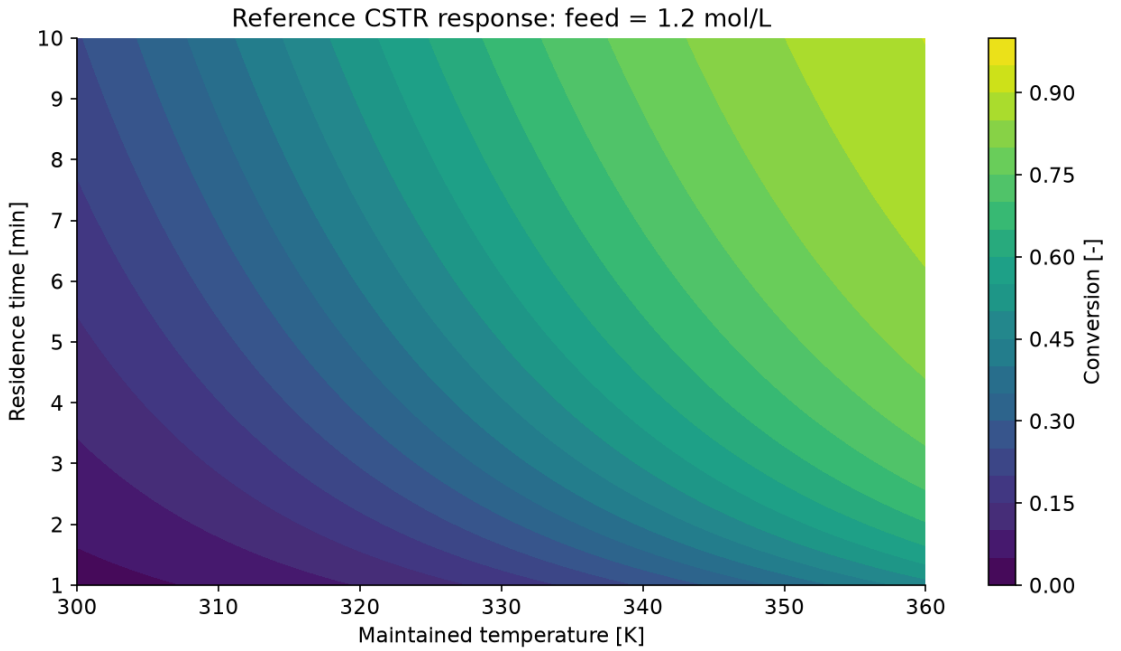
At 330 K, $\tau=5$ min, and $C_{Af}=1.2$ mol/L, $k=0.2$/min. Treat concentration as the unknown and rearrange the balance.

Step	Calculation	Interpretation
Zero accumulation	$0=(1.2-C_A)/5-0.2C_A$	Flow and reaction balance
Multiply by residence time	$0=1.2-C_A-1.0C_A$	Every term now in mol/L
Collect concentration	$2C_A=1.2$	Add coefficients of the unknown
Concentration and conversion	$C_A=0.6$; $X=1-0.6/1.2=0.5$	Half of A converted

Substitution gives a net flow term $(1.2-0.6)/5=0.12$ mol/(L·min) and reaction consumption $0.2\times0.6=0.12$. Their equality explains why concentration no longer changes.

Read the reference response surface

At 330 K and 5 min, conversion is 0.5; with feed 1.2 mol/L, outlet concentration is 0.6 mol/L.



At fixed feed, raising temperature or residence time increases kτ. The larger denominator lowers A concentration and raises conversion. Each point in the surface comes from evaluating these equations at another input.

4. Keep data roles separate

Independent samples within the declared box: 400 train, 120 validation, 160 test; seed 42.

Data roles: train for coefficient fitting; validation for model choices; test for the final error report. Supplied baseline: a fixed quadratic basis.

Error on points already used to choose coefficients does not measure error at new conditions. Repeated model selection also uses information from validation. A separate final test set therefore remains outside both fitting and selection.

5. Fit a small quadratic baseline

Scale each input using its declared lower and upper bounds:

$$\tilde{x}_j = 2\frac{x_j - l_j}{u_j - l_j} - 1$$

Ten features: a constant, three scaled inputs, three squares, and three pairwise products. Matrix dimensions: Φ 400 × 10; Y 400 × 2; B 10 × 2.

Coefficient fitting for both outputs: least squares

$$\min_B \|\Phi B - Y\|_F^2, \quad \hat{y} = \varphi\left(x\right)B$$

At prediction time, keep the fitted regression coefficients fixed and evaluate a new input.

Squares and cross-products represent curvature and interactions in the response surface. The predictor is quadratic in its inputs but linear in the unknown coefficients B, so fitting remains a linear least-squares problem.

Fitting least-squares coefficients to three points

Use dimensionless data x=[−1,0,1], y=[1,2,2] and model b+ax. No line passes through all three points, so choose coefficients that minimize the sum of squared errors.

For errors $e_i=b+ax_i-y_i$, differentiation gives $2\sum e_i=0$ for b and $2\sum x_i e_i=0$ for a. Substituting $\sum x_i=0$, $\sum x_i^2=2$, $\sum y_i=5$ and $\sum x_i y_i=1$ gives 3b=5 and 2a=1.

x	Target y	Prediction 5/3+x/2	Error
−1	1	7/6	1/6
0	2	5/3	−1/3
1	2	13/6	1/6

```
import numpy as np
x = np.array([-1., 0., 1.])
phi = np.column_stack((np.ones(3), x))
coef = np.linalg.lstsq(phi, [1., 2., 2.], rcond=None)[0]
prediction = phi @ coef  # [7/6, 5/3, 13/6]
```

coef=[b,a] and the squared-error sum is 1/6. The CSTR fit extends the feature matrix to ten columns and applies the same calculation to each output column. `lstsq` solves the least-squares problem without explicitly forming an inverse.

From one input row to ten features

Scaling [330, 5, 1.2] with the declared bounds gives [a,b,c]=[0,−1/9,1/7]. Inputs with different units and magnitudes move to the same −1 to 1 interval.

Feature group	Values for this row
Constant and inputs	1, 0, −0.111111, 0.142857
Squares a^2,b^2,c^2	0, 0.012346, 0.020408
Products ab,ac,bc	0, 0, −0.015873

```
phi = features(np.array([[330., 5., 1.2]]))
phi_train = features(train_x)
coef = np.linalg.lstsq(phi_train, train_y, rcond=None)[0]
prediction = phi @ coef
```

Shapes are (1,10), (10,2), and (1,2) for phi, coef, and prediction. Matrix multiplication weights and sums feature columns separately for each output. Preserve the training feature order at new inputs.

The residence-time column is 2×(5−1)/(10−1)−1=−1/9; the feed column is 2×(1.2−0.8)/(1.5−0.8)−1=1/7. Squared features square these transformed values, not the physical values 5 and 1.2.

6. Evaluate errors and output consistency

Per-output error on the held-out test set:

$$\mathrm{RMSE}_j = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\hat{y}_{nj} - y_{nj}\right)^2}$$

Concentration RMSE is in mol/L. Conversion RMSE is dimensionless; 0.025608 corresponds to about 2.56 percentage points. Also check whether the predicted pair agrees with the definition of conversion.

$$r = \hat{C}_A + C_{\mathrm{Af}}\hat{X} - C_{\mathrm{Af}}$$

The reference gives r = 0. An unconstrained two-output regression can disagree. A small average RMSE does not mean every condition has a small error.

Residual signs show overprediction and underprediction, but averaging signed residuals can cancel them. Squaring prevents that cancellation; the final square root restores the original output unit.

From residuals to RMSE and consistency

Compare reference concentrations [0.60,0.40] mol/L with predictions [0.65,0.38] mol/L.

Calculation	Result
Prediction minus reference	[0.05,−0.02] mol/L
Square	[0.0025,0.0004] (mol/L)²
Average	0.00145 (mol/L)²
Square root	RMSE≈0.038079 mol/L

```
error = prediction - truth
rmse = np.sqrt(np.mean(error**2, axis=0))
```

axis=0 averages input rows and retains output columns. Averaging both outputs together would mix concentration with dimensionless conversion. Separately, C_A=0.65, X=0.45, C_Af=1.2 gives 0.65+1.2×0.45−1.2=−0.01 mol/L. Accuracy and the relation between outputs need separate checks.

Prediction results to reproduce

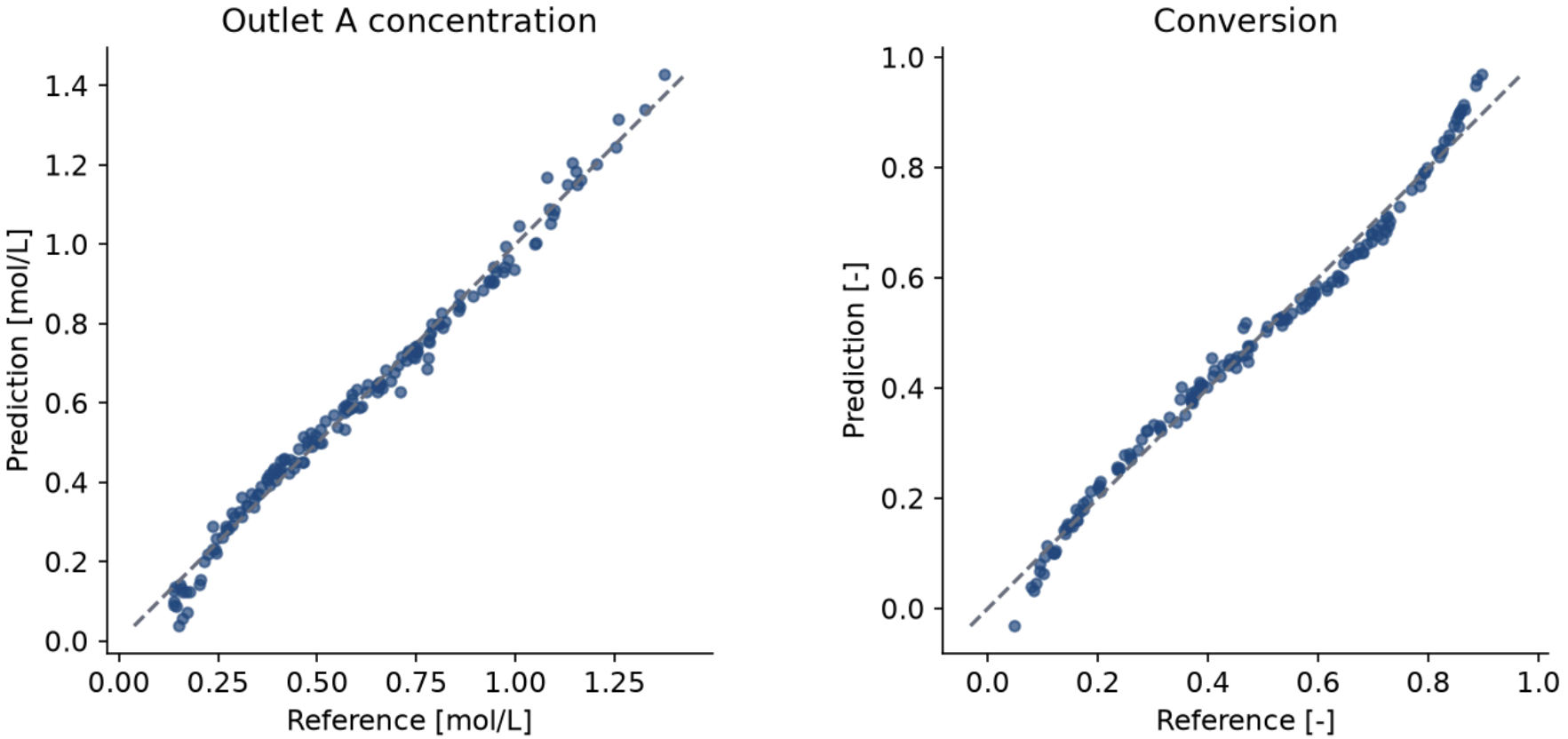
Check	Seed-42 baseline
Test concentration RMSE	0.032623 mol/L
Test conversion RMSE	0.025608
Test samples	160 independent points
Model	Least squares on ten quadratic features

Judge whether these errors are small enough for a particular use by inspecting parity plots and residuals together.

A conversion error of 0.025608 is about 2.56 percentage points on the 0–1 scale, not necessarily 2.56% relative error. Small average error can coexist with a large error at one condition; inspect the maximum and its input location as well.

Prediction error across operating conditions

The diagonal represents prediction equal to reference. Compare concentration and conversion separately, then relate the larger errors to the input conditions.



Points above the diagonal are overpredictions; points below it are underpredictions. Systematic curvature differs from random scatter. Plot residuals against temperature and residence time to locate regions where the quadratic basis misses response curvature.

7. Run the prediction lab

Download the notebook or script, plus requirements.txt.

```
python -m pip install -r requirements.txt
python week01_lab.py --output-dir week01-results --seed 42
```

The default run saves the dataset, prediction metrics, the reference map, and test parity plots. Compare the published dataset and recorded results.

Optional later-topic examples: retained in the script behind `--extensions`

8. Exercises and required submission

1. Derive C_A and X from the steady-state balance, with units.
2. Calculate the response at 330 K, 10 min, and feed 1.2 mol/L.
3. Count the quadratic features and state the dimensions of Φ, B, and Y.
4. Re-run with another seed and compare held-out errors. Explain why train error alone is insufficient.
5. For predicted $C_A = 0.65$ mol/L and $X = 0.45$ at feed 1.2 mol/L, calculate the consistency residual.

Submission: prediction specification, fitted baseline, and test report with units, parity plots, and consistency diagnostics.

Answer checks: $C_A = 0.4$ mol/L and $X = 2/3$ in Question 2; ten features; residual -0.01 mol/L in Question 5.

Reading and the next prediction task

Week 1 companion · English PDF

Using a predictor to select operating conditions requires objectives, operating constraints, and reference-model revalidation. Weeks 5-8 develop that process.